Introductions

What is the point of this class?

Course Administration

Criteria	Percentage
Attendance during final project presentations	5%
Projects (10% x 3)	30%
Final project	25%
Midterm exam	15%
Final exam	25%
Total	100%

Projects

- 3 projects, all individual
- Will lose 10%/day (or fraction thereof) of maximum grade (***I'll be a stickler for this too***)
  - That is 1% of your overall final grade in the course!
- Additional details will be given when projects are assigned

Final Project

- 5% of your grade will be for showing up ***on time*** to the final project presentations
  - 1% for showing up each day, 0.5% for being on time that day
  - Only 0.5% for being on time the day you're set to present

- Final project content will be based on selecting a dataset that is of interest to you, parsing the dataset, selecting the appropriate machine learning protocol to analyze the data, then presenting to the rest of the class

Exams

- Exams will generally focus more on concepts and methods rather than details (note: “focus more” does not mean “exclusively cover”)

- Examples of the types of questions that you’re likely to encounter:
  - Which machine learning method would be most appropriate for the following data
  - Computing/problem solving either with math or with pseudocode
  - Data analysis plan, what’s wrong with it?
  - Potentially fact-based questions about major topics

Lecture 1

COM 307: Machine Learning & Data Mining

What is machine learning?

- Scientific study of algorithms and statistical models that computer systems use to perform a specific task *without using explicit instructions*

- Generally agreed to be a subfield of artificial intelligence

- Generally use “training data” to make predictions on what a certain kind of data generally “looks like” to solve problems never encountered before

What is learning?

- Learning is generally a process of *generalization*

What is 1598274+4987?

How do you know? Have you added these two numbers before?

You know how to do this because you *learned* addition, so you can solve addition problems, even those that you've never encountered before!

What is data mining?

- Process of discovering patterns in large data sets, particularly at the intersections of machine learning, statistics, and database systems

- Interdisciplinary field of computer science and statistics with the goal of extracting information and determining structure of the data

Machine learning vs data mining

- They are very closely related, which is why they’re being covered together in this course

- Machine learning approaches work on 2 sets of data:
  - Training data used to determine variables in a data set
  - Test data, the stuff you actually care about
- Data mining focuses on the discovery of unknown properties of data

Machine learning vs data mining

- They are very closely related, which is why they’re being covered together in this course

- Machine learning tools are evaluated based on their ability to reproduce known knowledge effectively
- Data mining is focused on discovery of previously unknown knowledge

Data mining vs data analysis

- Data analysis is used to test models on a data set
  - Example: Analyze the effectiveness of a marketing campaign, regardless-ish of how much data you have

- Data mining finds hidden patterns in large volumes of data – don’t necessarily need to know what you’re looking for

Some examples—machine learning

- Email filtering (spam or not spam)
- Computer vision
  - Self-driving cars
  - Surveillance
  - Manufacturing/QC
  - …
- Agriculture
- Handwriting recognition
- Bioinformatics

Some examples—data mining

- Business
  - Walmart processes 20 million sales/day, who, what, and where?
  - Credit cards & store loyalty cards
  - Determining employee success
- Human rights, particularly related to the justice system
- Medical data mining (“genes” & diseases)
- Educational research
  - Student success
  - Student retention